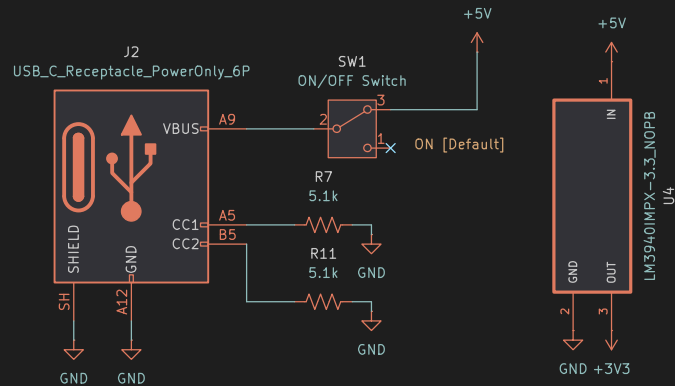


	1	2	3	4
A	Sheetname: Power Management Sets 5V and 3V3 Sheetfile: pwr_mgmt.kicad_sch	Sheetname: ESP Reciever BCKD DIND LRCLKD File: esp_rx.kicad_sch	Sheetname: DAC D>BCK D>DIN D>LRCLK DAC_OUT_LD DAC_OUT_RD File: dac.kicad_sch	
B	Sheetname: Volume Control D>DAC_OUT_L D>DAC_OUT_R POST_POT_IN_LD POST_POT_IN_RD SHDN_SIGNALD File: vo_ctrl.kicad_sch		Sheetname: Amp D>POST_POT_IN_L D>POST_POT_IN_R D>SHDN_SIGNAL AMP_OUT_LD AMP_OUT_RD File: amp.kicad_sch	
C			Dual Gang pot has a hard mute at 0% volume. DAC features a soft mute switch. Data: ESP -> DAC -> AMP-> 3.5mm Jack. damian m Sheet: / File: ESP32IEMCAD.kicad_sch Title: ESP32 Wireless IEM Rig Reciever Size: A5 Date: 2026-08-04 KiCad E.D.A. 10.0.5 Rev: 0.0.0 Id: 1/6	
	1	2	3	4



Using a low dropout regulator to step from 5V to 3V3 for ESP, DAC, and Amp.
 CC1, CC2 have a pull down resistor to GND to signify it is an Upstream Facing Port (UFP)
 Connect to a USB-C power bank. Will provide power to the entire RX circuit.
 Data: ESP -> DAC -> AMP-> 3.5mm Jack.

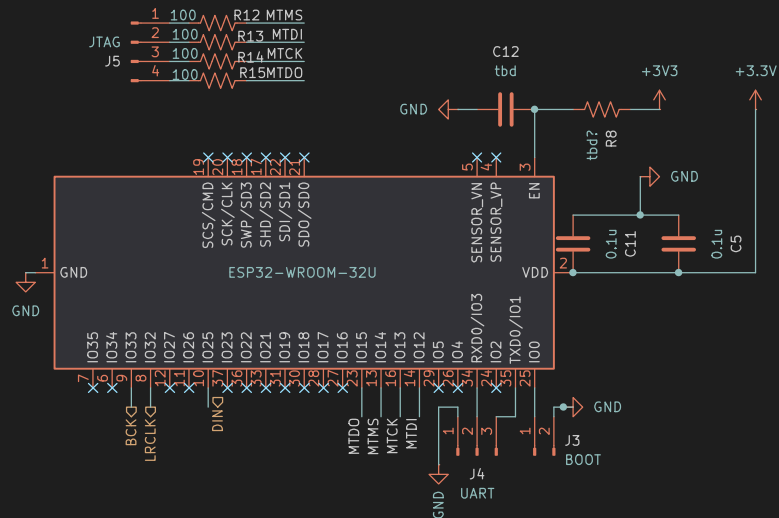
damian m

Sheet: /Power Management/
 File: pwr_mgmt.kicad_sch

Title: ESP32 Wireless IEM Rig Reciever

Size: A5 Date: 2026-08-04
 KiCad E.D.A. 10.0.5

Rev: 0.0.0
 Id: 2/6



Recieves the compressed data from ESP TX and decodes them. Then sends to DAC.
Data: ESP -> DAC -> AMP-> 3.5mm Jack.

damian m

Sheet: /ESP Reciever/

File: esp_rx.kicad_sch

Title: ESP32 Wireless IEM Rig Reciever

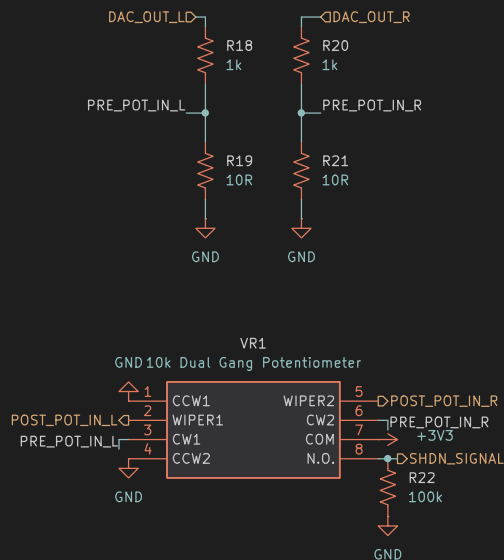
Size: A5

Date: 2026-08-04

Rev: 0.0.0

KiCad E.D.A. 10.0.5

Id: 3/6



POST_POT_IN_R/L becomes the input to the amp.
 Volume Knob with integrated off switch to kill amp from playing any audio.
 AMP_IN_L/R follows a voltage divider to lower volume to safer levels before hitting AMP.
 Data: ESP -> DAC -> AMP-> 3.5mm Jack.

damian m

Sheet: /Volume Control/

File: vol_ctrl.kicad_sch

Title: ESP32 Wireless IEM Rig Receiver

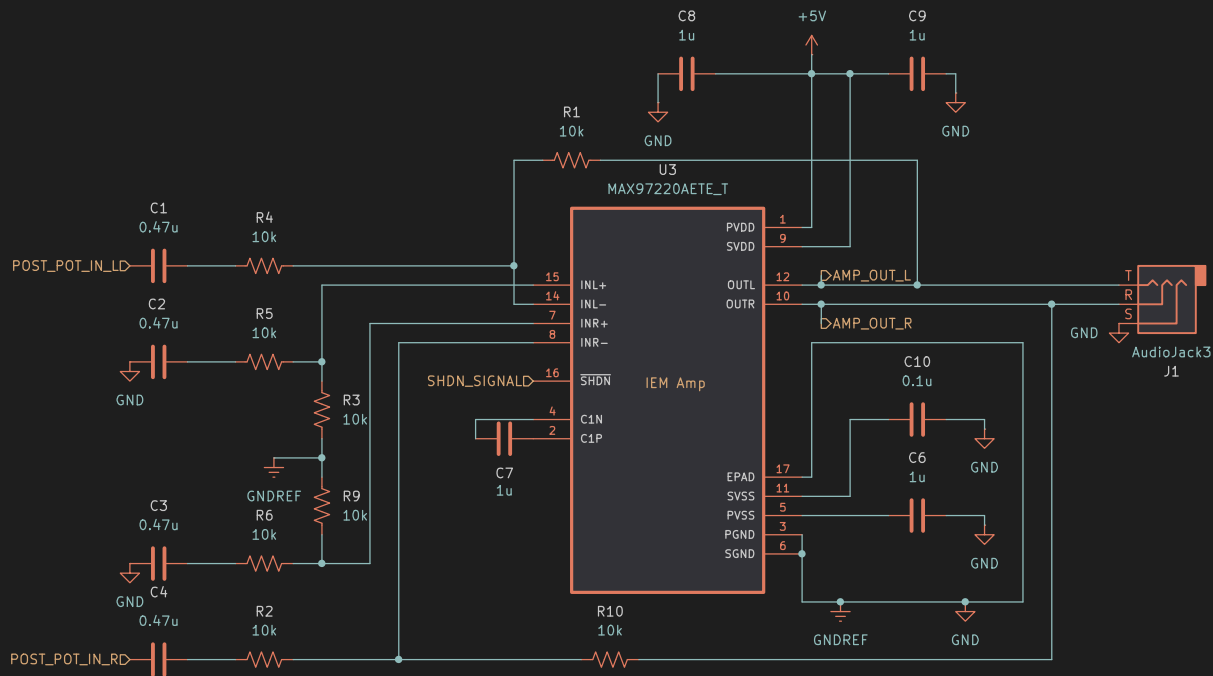
Size: A5

Date: 2026-08-04

Rev: 0.0.0

KiCad E.D.A. 10.0.5

Id: 5/6



SHDN_SIGNAL LOW acts as setting the volume to a "true 0%" by shutting off the IEM amp.

This is in contrast to the DAC Soft mute which acts as a mute toggle without changing the volume.

Amp follows the pseudo-differential input configuration because the DAC output is a single-ended input.

SHDN_SIGNAL is handled by the dual gang pot. N.O. = AMP OFF.
BIAS is in the same position as SVSS. Refer to in manual as BIAS.
Data: ESP -> DAC -> AMP-> 3.5mm Jack.

damian m

Sheet: /Amp/
File: amp.kicad_sch

Title: ESP32 Wireless IEM Rig Reciever

Size: A5 Date: 2026-08-04
KiCad E.D.A. 10.0.5

Rev: 0.0.0
Id: 6/6